



Figure. The EPSS-TPG model at a glance. The top row is the *graph branch*. Starting from the raw CVE description text, the spaCy frontend together with a regex and dictionary security pass builds a Text Property Graph; SecBERT then attaches a 768-dimensional contextual embedding to every node, and the result is packed into a PyG graph with N nodes carrying 781-dimensional features (a 13-way one-hot node-type vector concatenated with the SecBERT embedding) and 13 base edge types, or 23 once the security view is enabled. A multi-view gated GNN encodes the graph one edge family at a time and combines the per-view embeddings by attention; a mean || max pool then flattens everything into a 256-dimensional graph vector $g \in \mathbb{R}^{256}$. The bottom row is the *tabular branch*. Fifty-five hand-crafted CVE features (CVSS sub-scores, CWE indicators, age, exploit flag and reference counts) pass through a small MLP that emits a 64-dimensional vector $t \in \mathbb{R}^{64}$. The two branches meet at the fusion head, which concatenates g and t into a 320-dimensional vector, runs it through a $320 \rightarrow 128 \rightarrow 64 \rightarrow 1$ MLP, and applies a sigmoid to read off the predicted exploitation probability $\hat{p} = \sigma(\hat{\ell})$.